

Landscape Cinemagraph Synthesis with Sketch Guidance

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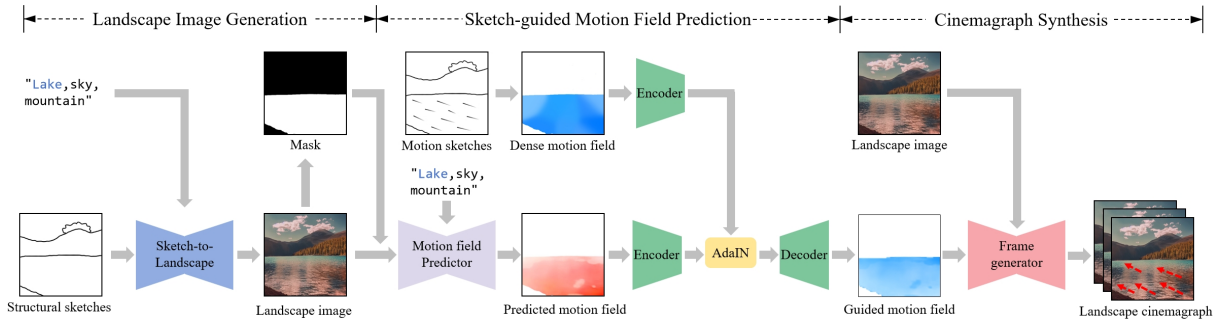


Figure 1: Given input hand-drawn sketches of landscape structure and motions, the proposed framework can generate landscape cinemagraphs with landscape image generation, sketch-guided motion field prediction, and cinemagraph synthesis steps.

CCS Concepts

• Computing methodologies → Computer vision.

Keywords

Cinemagraphs synthesis, user sketches, generation models

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1 Introduction

Cinemagraphs embody a captivating fusion of photography and video, animating parts of still images to create a striking visual effect. However, it is challenging to create visually pleasing cinemagraphs that require a significant investment of time and skills, especially for common users without professional knowledge and skills. In computer graphics, physics-based simulations are used

to infer the motion trajectories of fluid elements which are time-consuming and domain-specific. Recently, deep learning based diffusion models can generate high-quality images and videos. Text2Cinemagraph can generate a realistic and artistic cinemagraph from a text prompt [Mahapatra et al. 2023]. The input modalities of these methods are only still real-world images or conditioning with text prompts which are difficult for common users to generate and control the landscape cinemagraph with specific design intentions.

To solve these issues, we present a novel landscape cinemagraph synthesis framework with sketch-guided approach from freehand sketches using diffusion models and motion predictor as shown in Fig. 1. We propose a fine-tuned latent diffusion model to generate realistic fluid elements, such as waterfalls and sea, from the provided structural sketches and a deep-fusion method to adjust the predicted motion field based on motion sketches.

2 Methods

Fig. 1 illustrates the proposed framework for cinemagraph synthesis using sketch guidance. First, the landscape image generation step adopts latent diffusion model (LDM) to generate a realistic landscape image from structural sketch and text. However, the baseline LDM model has limitations in accurately generating naturalistic outdoor scenery due to an adequate quantity of high-quality landscape images in training data (Fig. 2b). Therefore, we conduct the fine-tuning of LDM model on a real landscape dataset [Holynski et al. 2021]. Then, the pre-trained ControlNet is adopted to control

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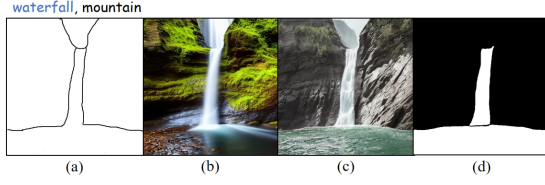


Figure 2: (a) Structural sketches; landscape images generated by LDM model (b) and fine-tuned LDM model (c), and fluid masks extracted by Grounded-SAM (d).

the spatial regions and contents during the image generation process. To generate dynamic fluid elements while keeping background still, we extract the fluid mask from the generated image using the Grounded-SAM method [Ren et al. 2024](Fig. 2d).

The sketch-guided motion field prediction step aims to involve the fluid elements in the generated image with motion dynamics. A motion field is predicted from the generated realistic landscape image I_0 , using a motion predictor conditioned with text prompt t and mask M , trained on the landscape dataset. To align the predicted motion field with the given motion sketches, we adopt the distance-based method [Mahapatra and Kulkarni 2022] to convert motion sketches to dense motion field. In contrast to previous work, we propose a deep-fusion method to adjust the predicted motion field using dense motion field in latent space. Specifically, we train a CNN-based motion field autoencoder to compress predicted and dense motion fields into latent space. The predicted motion field is transformed to align with the dense field’s distribution for motion fusion, and then decoded to animate the still image.

The cinemagraph synthesis step uses the guided motion field to compute the 2D position of each pixel in the n -th frame to the $(n+1)$ -th frame. The seamless looping cinemagraph can be synthesized with symmetric splatting in latent space. The displacement fields are applied to bidirectionally warp the deep feature map D_0 to D_{-n} and D_n , and a softmax function is utilized to determine the contributions of colliding source pixels in the target frame. Finally, the encoded frame set is decoded to generate the final output cinemagraph.

3 Results

Fig.3 shows the generated results of landscape image and motion fields from the given sketches. The input sketches consist of structural sketches and motion sketches (Fig.3a) where the directions of motion sketches are represented by gradient color from white to black. The fine-tuned LDM model can generate realistic landscape image from the structural sketches part, which produces natural fluid surface characteristics such as splashes, ripples and reflections (Fig.3b). The predicted motion field (Fig. 3c), dense field (Fig. 3d) and guided field (Fig. 3e) demonstrate that our deep-fusion method can retain the distribution of the original predicted motion field while aligning with the direction of the user-provided motion sketches. It is verified that the proposed framework can generate the visually pleasing landscape cinemagraphs directly from freehand sketches. More details are in supplementary video.

Table 1 shows the comparison results of our framework with *Text2Cinemagraph*. We evaluated the motion quality by measuring the average PSNR and MS-SSIM between the predicted motion

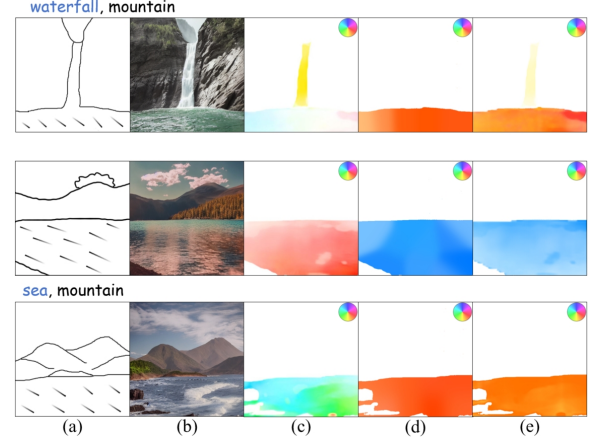


Figure 3: User sketches (a) are used to generate landscape image (b) and dense motion field (d), then the motion field (c) is predicted to obtain the final sketch guided field (e).

Table 1: Comparison results with previous work.

Method	Motion Quality		Content Consistency		
	PSNR $\uparrow$	MS-SSIM $\uparrow$	LPIPS $\downarrow$	MS-SSIM $\uparrow$	RMSE $\downarrow$
Text2Cine.	12.6484	0.7070	0.0541	0.8094	0.0716
Ours	17.7545	0.7845	0.0470	0.8110	0.0672

field and the guided field. Our framework achieved higher scores, showing it can effectively preserve the distribution of the original predicted motion field while adapting to the direction of the sketches. The content consistency is assessed by calculating the average LPIPS, MS-SSIM and RMSE between the first frames and the subsequent frames. The results show that our framework can generate motions that better preserve landscape contents.

4 Conclusions

This paper proposes a sketch-guided approach for generating landscape cinemagraphs from freehand sketches. The proposed approach can generate visually pleasing landscape cinemagraphs from the provided structural and motion sketches. The generation of landscape cinemagraphs directly from freehand sketches can greatly reduce the complexity of the creation process. This enables the common users without design training and skills to create landscape cinemagraphs with intuitive input.

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